

Data Cleaning Log

Project: Olist Brazilian E-Commerce Dataset

This document records all preprocessing and data cleaning activities performed on the Olist Brazilian E-Commerce dataset. The objective was to improve data quality, handle missing values, remove duplicate records, standardize the data, and prepare the datasets for exploratory data analysis and predictive modeling.

1. Duplicate Record Removal

Dataset Affected: Geolocation Dataset

Duplicate geographical records were identified and removed using the `drop_duplicates()` function. This eliminated redundant information while preserving unique records.

2. Handling Missing Values

Dataset Affected: Order Reviews Dataset

Missing values in `review_comment_title` and `review_comment_message` were replaced with 'No Title' and 'No Review'. This preserved all customer reviews without deleting observations.

3. Handling Missing Values

Dataset Affected: Products Dataset

Missing product categories were replaced with 'Unknown'. Missing numerical attributes such as dimensions, weight, and description length were filled using the median value of each respective column.

4. Data Type Conversion

Dataset Affected: Orders, Order Reviews and Order Items

All date-related columns were converted from object to datetime format, including `shipping_limit_date`, enabling accurate time-based analysis.

5. Text Standardization

Dataset Affected: Multiple Datasets

Leading and trailing white spaces were removed from selected text columns in the Customers, Sellers, Products, and Geolocation datasets to improve consistency.

6. Data Validation

Dataset Affected: All Datasets

Validation was performed by checking missing values, duplicate records, dataset dimensions, and data types after cleaning.

7. Business Rule Validation

Dataset Affected: Orders Dataset

Remaining missing values in order_approved_at, order_delivered_carrier_date, and order_delivered_customer_date were intentionally retained because they correspond to cancelled, unavailable, or undelivered orders.

Cleaning Log Summary

Step	Dataset	Cleaning Operation	Status
1	Geolocation	Removed duplicate records	Completed
2	Order Reviews	Handled missing values	Completed
3	Products	Handled missing values	Completed
4	Orders, Order Reviews, Order Items	Converted date columns to datetime	Completed
5	Multiple Datasets	Trimmed white spaces	Completed
6	All Datasets	Validated cleaned datasets	Completed
7	Orders	Retained business-related missing values	Completed

Conclusion

The data cleaning process successfully improved the quality, consistency, and reliability of the datasets. Duplicate records were removed, missing values were handled appropriately, date columns were converted to datetime format, text fields were standardized, and the cleaned datasets were validated. The remaining missing values in the Orders dataset were retained because they represent meaningful business events rather than data quality issues.